Daattavya **Aggarwal**

PhD in AI for Mathematical Discovery, University of Cambridge

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Interested in AI guided mathematical discovery

Research Interests

- Developing methods for AI guided maths, with an emphasis on discovering new concepts and conjectures and applications in active research problems.
- Previous work in topology, geometry, number theory and quantum computing.

Education

PhD in AI for Mathematical Discovery *Jan. 2024 – ~ Summer 2027*

University of Cambridge, Cambridge, UK

Supervisor: Dr. Challenger Mishra

MSc. Mathematical and Theoretical Physics *Oct. 2020 - Jun. 2021*

Mathematical Institute, University of Oxford, Oxford, UK

Pass with **Merit**, **Distinction** in Thesis

B.Tech in Engineering Physics *July 2016 - May 2020*

Indian Institute of Technology, Roorkee, India

CGPA: 8.738/10, **First Division with Distinction**

Selected Publications and Preprints

[1] **Aggarwal, D.**, Kim, O., Ek, H. C. & Mishra, C. (2026) "Discovering mathematical concpets through a multi-agent system" *arXiv preprint:* arXiv:2603.04528

[2] Mirjanić, V., **Aggarwal, D.** & Mishra, C. (2025) "A remark on weighted average multiplicities in prime factorisation" *arXiv preprint:* arXiv.2510.06993

[3] **Aggarwal, D.**, He, Y.-H., Heyes, E., Hirst, E., Sá Earp, H. N. & Silva, T. S.R. (2024). "Machine Learning Sasakian and G_2 topology on contact Calabi-Yau 7-manifolds". *Physics Letters B*, 850, 138517. doi: 10.1016/j.physletb.2024.138517

[4] Gangopadhyay, S., Poonia, V. S., **Aggarwal, D.** & Parekh, R. (2019). "Generalized Boolean Functions and Quantum Circuits on IBM-Q". *ICCCNT 2019, pp. 1-6.* doi: 10.1109/ICCCNT45670.2019.8944437

[5] **Aggarwal, D.**, Raj, S., Behera, B.K. & Panigrahi, P.K. (2018). "Application of quantum scrambling in Rydberg atom on IBM quantum computer". *arXiv preprint:.* arXiv.1806.00781

Conferences and Talks

Apr. 2026	DANGER: Data, Numbers and Geometry - Talk on “Discovering mathematical concepts through a multi-agent system”	*BIRS, Banff*
Feb. 2026	AIxMathematics: Talk on “Discovering mathematical concepts through a multi-agent system”	*ICMS, Edinburgh*
Oct. 2025	ML retreat: Talk on “Data-driven mathematics”	*University of Cambridge*
Aug. 2025	Attended workshop: AI Meets Mathematics	*Imperial College London*
Sep. 2024	Attended conference: Geometry from the model theorist’s point of view	*University of Oxford*
Feb. 2024	Wolfson XML: “AI for Mathematical Discovery”	*Wolfson College, Oxford*
Dec. 2023	Poster on “Machine Learning Sasakian and G_2 topology on contact Calabi-Yau 7-manifolds” at neurips@camb	*University of Cambridge*
Oct. 2023	Talk on “Machine Learning G_2 manifolds”	*Computer lab, Cambridge*
Jan. 2019	Poster on “Boolean Functions for Quantum Circuits” at IndoQuant	*IIT Hyderabad*

Teaching & Supervision

2023-2024	Supervised an MPhil in Advanced Computer Science dissertation: Deepro Choudhary	*University of Cambridge*
2026	Led supervisions for IB: Prolog	*University of Cambridge*
2025	Led supervisions for IA: Machine Learning and Real-World Data	*University of Cambridge*
2024	Led supervisions for IB: Artificial Intelligence	*University of Cambridge*

Academic Achievements & Grants

2023	UKRI Grant: Covering my position as a research assistant and PhD student at the Computer Lab	*University of Cambridge*
2020	Felix Scholar: Awarded a full scholarship for postgraduate study	*University of Oxford*

Professional Appointments

Sep 2023 - Sep 2026	Research assistant: Department of Computer Science and Technology	*University of Cambridge*
Jan 2023 - Aug 2023	Junior Data Scientist: Machine learning for digital biomarkers in neurology	*Kneu Health, London*
Aug 2022 - Dec 2022	Research Data Scientist: Parkinson’s Disease Diagnostics	*Oxford Parkinson Disease Centre*
Dec 2019 - Jan 2020	Visiting Research Assistant: Adiabatic Quantum Computation	*Information Sciences Institute, USC*

Extracurricular Activities

2025 - Present	MCR Treasurer, Girton College: Representing the postgraduate student body	*University of Cambridge*
Apr 2021	Team Captain, Wolfson Oxford: Represented the college on the BBC quizzing show, *University Challenge*.	*Salford, UK*
May 2017 - Mar 2018	Member, Team Robocon IITR: Won “Best Innovative Design” and finished 7th/107 at ABU Robocon	*Roorkee, India*